

Emran Masoumifeshani, PhD

Sr. AI Data Scientist | AI/ML Engineer

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Profile

PhD-trained scientist and AI/ML engineer with hands-on experience building Generative AI applications, agentic systems (LangChain, LangGraph), LLM evaluation frameworks, and model lifecycle governance, complemented by a strong background in computational modeling, statistical analysis, and data engineering (GCP, Databricks, SQL/NoSQL). Brings 7+ years of professional experience across industrial R&D, scientific computing, and applied AI/data engineering.

Proven track record designing scalable computational methodologies and delivering reproducible, data-driven solutions for large-scale scientific challenges – including high-performance computing and cross-functional collaboration – resulting in publications in leading journals such as *Angewandte Chemie*, *Journal of Chemical Theory and Computation*, and *Journal of Computational Chemistry*. Skilled in explainable AI (SHAP), bias and fairness assessment, and regulatory compliance (GDPR): well suited to AI engineering, Model Risk Management, and applied AI roles within highly regulated industries such as financial services.

Core Skills

- **AI/ML & Model Risk Management (MRM):** Generative AI & LLMs (LangChain, LangGraph), Independent Model Validation (IMV), LLM Evaluation (LangSmith, DeepEval), LLMOps, Model Lifecycle Governance, Explainability (SHAP), Bias & Fairness, GDPR / Regulatory Compliance
- **Agentic AI Systems:** Design and evaluation of agent-based architectures and autonomous decision frameworks
- **Programming:** Python, C++, R, JavaScript
- **Data & Cloud:** GCP BigQuery, Databricks, SQL, NoSQL (MongoDB), Spark, PySpark, ETL, Data Modeling, Vector Databases, Azure, AWS
- **ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, Model Versioning (MLflow)
- **Analytics & Experimentation:** A/B Testing, Causal Analysis, Statistical Modeling
- **Data Platforms & BI:** Power BI, Tableau, Looker Studio
- **Engineering & MLOps:** FastAPI, Airflow, Docker, CI/CD, Git, API Testing (Postman), JIRA, Confluence

Professional Experience

Founder | Sr. AI Data Scientist | AI/ML Engineer

Feb 2025 – Present

Royal Dywany, Warsaw

- Led design, development and evaluation of agentic AI architectures using LangChain and LangGraph to automate decision workflows and business processes, cutting manual turnaround time.
- Built and validated LLM evaluation pipelines (LangSmith, DeepEval) covering accuracy, robustness, explainability (SHAP), and bias/fairness assessment prior to deployment across two business-facing AI applications.
- Directed the company's data platform strategy across GCP BigQuery and Databricks, including ETL pipeline design, data modeling, and integration of SQL/NoSQL (MongoDB) and vector databases for retrieval-augmented generation (RAG).
- Implemented core MLOps practices including model versioning, Docker-based deployment, API development and CI/CD workflows.
- Implemented model lifecycle governance and GDPR-aligned data compliance processes, and introduced BI reporting (Power BI, Looker Studio) for stakeholder decision-making.
- Managed a cross-functional engineering team of 5 and delivery workflow using Git, JIRA and Confluence, overseeing product strategy alongside technical execution.

Sr. Research Assistant & Assistant Professor

The Institute of Organic Chemistry, Polish Academy of Sciences (IOC PAS), Warsaw

Sep 2023 – May 2025

- Conducted advanced computational research involving large-scale scientific datasets, predictive modelling, statistical analysis, and validation of computational models.
- Designed and implemented Python-based computational workflows for **data preprocessing, feature extraction, automated data generation, simulation pipelines, and reproducible computational experiments**.
- Applied rigorous **machine learning model evaluation and benchmarking principles** to assess predictive accuracy, robustness, uncertainty, reliability, and generalisation of computational models.
- Developed **data-driven and scientific machine learning approaches** for analysing complex molecular systems, integrating numerical simulations, statistical modelling, feature-based analysis, and quantitative prediction.
- Managed computational experiments through reproducible workflows, systematic validation, model benchmarking, and structured documentation aligned with **machine learning model lifecycle and reproducibility principles**.
- Collaborated with interdisciplinary research teams, translating complex computational and **data-driven modelling** results into actionable scientific insights and contributing to peer-reviewed publications.

Doctoral Researcher

University of Warsaw, Warsaw

Dec 2018 – May 2023

- Designed and developed the **AROFrag** fragmentation methodology for systematic decomposition of aromatic molecular systems, improving computational efficiency and accuracy of interaction modelling.
- Developed Python-based automated pipelines for **data generation, preprocessing, feature extraction, computational experiments, statistical analysis, model benchmarking, and quantitative result interpretation**.
- Performed extensive **model validation, performance evaluation, and benchmarking** using Density Functional Theory (DFT) and *ab initio* methods, applying quantitative metrics to assess model accuracy, robustness, uncertainty, and **generalisation capability**.
- Applied **statistical modelling, uncertainty quantification, quantitative data analysis, and hypothesis-driven evaluation** to investigate intermolecular interactions in supramolecular and nanomaterial systems.
- Served as a **Teaching Assistant for Computational Chemistry and Quantum Chemistry courses**, supporting students in computational methods, quantitative problem-solving, statistical analysis, and interpretation of computational results.
- Analysed **high-dimensional simulation and experimental datasets** using statistical and computational methods, identifying patterns, quantifying relationships, and translating complex data into scientifically meaningful conclusions.
- Designed reproducible computational workflows using **version-controlled environments, automated pipelines, systematic validation, and structured research processes** to ensure transparency, reliability, and reproducibility of scientific analyses.
- Communicated complex **computational, statistical, and data-driven modelling** results through peer-reviewed scientific publications and international conference presentations.

Chemistry Teacher

British International School, Warsaw

Nov 2019 – Jan 2020

- Delivered chemistry lessons in English within an international curriculum; planned and adapted engaging lessons for a diverse student body.
- Assessed student progress and maintained effective classroom management in a short-term teaching role.

Research & Development Engineer

Nirou Trans Company, Shiraz, Iran

Nov 2013 – Nov 2018

- Evaluated and qualified raw material applications for polymer and composite systems, assessing composition and properties against technical/safety data sheets in coordination with procurement, engineering and quality teams.
- Conducted chemical resistance, mechanical (tensile, elongation, flexural, hardness) and rheological/viscoelastic characterisation of polymer and elastomer systems.
- Executed accelerated ageing and thermal endurance studies under defined stress conditions, tracking material property changes over time.
- Developed and validated new test methods, drafting test programmes, acceptance criteria and SOPs for laboratory use.

Education

Doctor of Philosophy (Ph.D.) in Chemical Sciences

2018–2023

University of Warsaw, *Warsaw, Poland*

- Dissertation: Developed a novel computational methodology for systematic molecular fragmentation of branched molecules and polycyclic aromatic hydrocarbons, enabling scalable quantum chemical modeling of large molecular systems.

Master of Science (M.Sc.) in Physical Chemistry

2011–2013

University of Tehran, *Tehran, Iran*

- GPA: **18.66/20** **Ranked 1st in Class**
- Thesis: Determination of Potential Energy Curves for OH, CH⁺, and NeH⁺ Molecules Using Spectroscopic Data

Bachelor of Science (B.Sc.) in Chemistry

2007–2011

University of Tehran, *Tehran, Iran*

- GPA: **17.41/20** **Ranked 1st in Class**

Publications

- A. Pareek, M. Morawiak, **E. Masoumifeshani**, P. Gawel, “Photostable π -Expanded Furans via Base-Mediated Quinone Oxygen-Annulation,” *The Journal of Organic Chemistry*, 2026, **91**, 2837–2843. doi:10.1021/acs.joc.5c02906
- M. Y. Mehboob, **E. Masoumifeshani**, Z. Badri, C. Foroutan-Nejad, “Design of Carbon–Carbon Ylides,” *Angewandte Chemie International Edition*, 2025, **64**, e202507446. doi:10.1002/anie.202507446
- K. Bartkowski, **E. Masoumifeshani**, M. Kotowska, U. Klimczak, B. Furman, C. Foroutan-Nejad, M. Lindner, “One-Pot Transition-Metal-Free Synthesis of π -Extended Bipolar Polyaromatic Hydrocarbons,” *Angewandte Chemie International Edition*, 2025, **64**, e202423282. doi:10.1002/anie.202423282
- **E. Masoumifeshani**, T. Korona, “Intermolecular Interaction Energies with AROFRAG—A Systematic Approach for Fragmentation of Aromatic Molecules,” *Journal of Computational Chemistry*, 2024, **45**(29), 2446–2458. doi:10.1002/jcc.27429
- **E. Masoumifeshani**, T. Korona, “AROFRAG—A Systematic Approach for Fragmentation of Aromatic Molecules,” *Journal of Chemical Theory and Computation*, 2024, **20**, 1078–1087. doi:10.1021/acs.jctc.3c00875
- T. Korona, J. Jankowska, **E. Masoumifeshani**, “Dicarbon Defect in Hexagonal Boron Nitride Monolayer—A Theoretical Study,” *Canadian Journal of Chemistry*, 2023. doi:10.1139/cjc-2022-0291
- **E. Masoumifeshani**, M. Chojecki, D. Rutkowska-Zbik, T. Korona, “Association Complexes of Calix[6]arenes with Amino Acids Explained by Energy Partitioning Methods,” *Molecules*, 2022, **27**, 7938. doi:10.3390/molecules27227938
- **E. Masoumifeshani**, M. Chojecki, T. Korona, “Electronic Correlation Contribution to the Intermolecular Interaction Energy from Symmetrized Systematic Molecular Fragmentation Model,” *Computational and Theoretical Chemistry*, 2022, **1211**, 113684. doi:10.1016/j.comptc.2022.113684
- **E. Masoumifeshani**, T. Korona, “Symmetrized Systematic Molecular Fragmentation Model and Its Application for Molecular Properties,” *Computational and Theoretical Chemistry*, 2021, **1202**, 113303. doi:10.1016/j.comptc.2021.113303

Certifications

- Building Generative AI-Powered Applications with Python – IBM
- Generative AI Language Modeling with Transformers – IBM
- Gen AI Foundational Models for NLP & Language Understanding –IBM
- Generative AI and LLMs: Architecture and Data Preparation –IBM
- The Structured Query Language (SQL) – University of Colorado Boulder
- Foundations: Data, Data, Everywhere – Google